

Technical Note: Sieve-Orthogonalized Mean Empirical Likelihood

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1 Purpose and Model

This note records the proof of the Wilks result for a mean empirical likelihood statistic when the predictive regression contains an unknown exogenous-control function. The model is

$$Y_t = m(W_{t-1}) + \beta X_{t-1} + U_t, \quad X_t = \theta + \rho_T X_{t-1} + \varepsilon_t, \quad (1)$$

where $W_{t-1} \in \mathbb{R}^d$ is a stationary exogenous covariate vector and X_{t-1} is persistent. The maintained endogeneity structure is the CCHT-type predictive-regression setup; for example, one may write $U_t = \phi \varepsilon_t + z_t$ with the martingale condition imposed below.

The object of inference is β . The nuisance function m is approximated by an increasing-dimensional sieve and profiled out under the null value β_0 . The proof separates the imported CCHT oracle limit from the new sieve perturbation argument.

2 Notation and Profile Score

Let

$$\mathbf{Y} = (Y_1, \dots, Y_T)', \quad \mathbf{X}_{lag} = (X_0, \dots, X_{T-1})'.$$

Let $\mathbf{P}$ be the $T \times K$ matrix with row t equal to $\mathbf{P}_K(W_{t-1})'$. Define

$$\mathbf{\Pi}_K = \mathbf{P}(\mathbf{P}'\mathbf{P})^{-1}\mathbf{P}', \quad \mathbf{M}_K = \mathbf{I}_T - \mathbf{\Pi}_K.$$

For a candidate β , set

$$\hat{\mathbf{c}}_K(\beta) = (\mathbf{P}'\mathbf{P})^{-1}\mathbf{P}'(\mathbf{Y} - \beta\mathbf{X}_{lag}), \quad \hat{\mathbf{u}}(\beta) = \mathbf{M}_K(\mathbf{Y} - \beta\mathbf{X}_{lag}). \quad (2)$$

Let $w_t = w(X_{t-1})$, $\mathbf{w} = (w_1, \dots, w_T)'$, $\bar{w}_T = T^{-1} \sum_t w_t$, and $\tilde{w}_t = w_t - \bar{w}_T$. The sieve-orthogonalized weight and score are

$$\mathbf{w}^c = \mathbf{M}_K \mathbf{w}, \quad \hat{Z}_t(\beta) = \hat{u}_t(\beta) w_t^c. \quad (3)$$

For any vector $\mathbf{v} = (v_1, \dots, v_T)'$, write the empirical norm as $\|\mathbf{v}\|_T^2 = T^{-1} \sum_{t=1}^T v_t^2$.

The empirical likelihood ratio is

$$\ell_T(\beta) = 2 \sum_{t=1}^T \log\{1 + \hat{\lambda}(\beta) \hat{Z}_t(\beta)\}, \quad (4)$$

where $\hat{\lambda}(\beta)$ solves

$$\sum_{t=1}^T \frac{\hat{Z}_t(\beta)}{1 + \hat{\lambda}(\beta) \hat{Z}_t(\beta)} = 0. \quad (5)$$

3 Assumptions

Assumption 1 (Oracle CCHT mean-score conditions). *With respect to*

$$\mathcal{H}_t = \sigma\{(z_s, \varepsilon_s) : s \leq t\} \vee \sigma\{W_s : s \leq t\},$$

$E(U_t \mid \mathcal{H}_{t-1}) = 0$, $\sup_t E|U_t|^{2+\delta} < \infty$ for some $\delta > 0$, the persistence conditions of the CCHT mean-score theorem hold, the raw weight is uniformly bounded and satisfies the CCHT saturation condition, and the CCHT conditional-variance conditions hold. For

$$A_T = T^{-1/2} \sum_{t=1}^T U_t \tilde{w}_t, \quad V_T = T^{-1} \sum_{t=1}^T U_t^2 \tilde{w}_t^2,$$

there exists an a.s. positive, possibly random, η^2 such that

$$A_T \xrightarrow{\text{st}} MN(0, \eta^2), \quad V_T \xrightarrow{p} \eta^2, \quad \max_{t \leq T} |U_t \tilde{w}_t| = o_p(\sqrt{T}). \quad (6)$$

Assumption 2 (Sieve projection conditions). *The first component of $\mathbf{P}_K(W)$ is one. After centering the remaining basis functions in sample, write the associated matrix as $\mathbf{R}$, so $\mathbf{1}_T' \mathbf{R} = \mathbf{0}$ and $\text{span}(\mathbf{P}) = \text{span}(\mathbf{1}_T, \mathbf{R})$. The population timing restriction is*

$$\mathbb{E}[\mathbf{R}_K(W_{t-1})\{w(X_{t-1}) - \mathbb{E}w(X_{t-1})\}] = \mathbf{0}.$$

The eigenvalues of $T^{-1} \mathbf{R}' \mathbf{R}$ are bounded away from zero and infinity with probability tending to one. For $\zeta_K = \max_{t \leq T} \|\mathbf{R}_t\|$,

$$\|T^{-1} \mathbf{R}' \tilde{\mathbf{w}}\| = O_p(\sqrt{K/T}), \quad (7)$$

$$\|T^{-1/2} \mathbf{R}' \mathbf{U}\| = O_p(\sqrt{K}), \quad (8)$$

$$\|T^{-1} \mathbf{P}' \mathbf{U}\| = O_p(\sqrt{K/T}). \quad (9)$$

These sample projection bounds can be verified under the stated sieve orthogonality, exponential mixing of W_t , uniformly bounded sieve-basis moments, and the martingale condition in Assumption 1.

Assumption 3 (Approximation and growth). *For some $\mathbf{c}_K$,*

$$m(W_{t-1}) = \mathbf{P}_K(W_{t-1})' \mathbf{c}_K + r_{K,t},$$

where $\|\mathbf{r}_K\|_T = O(a_K)$, $\|\mathbf{r}_K\|_\infty = O(a_K)$, and $\sqrt{T}a_K \rightarrow 0$. The sieve projector is stable on the approximation residual: $\|\mathbf{M}_K \mathbf{r}_K\|_\infty = O(a_K)$. Also,

$$K/\sqrt{T} \rightarrow 0, \quad \zeta_K \sqrt{K/T} \rightarrow 0.$$

If $a_K \asymp K^{-s/d}$, a sufficient window is $T^{d/(2s)} \ll K \ll T^{1/2}$, nonempty when $s > d$.

Assumption 4 (Nondegeneracy and convex hull). *There exists $\underline{\eta} > 0$ such that $P(\eta^2 \geq \underline{\eta}^2) = 1$. Under $H_0 : \beta = \beta_0$,*

$$P\{\min_{t \leq T} \hat{Z}_t(\beta_0) < 0 < \max_{t \leq T} \hat{Z}_t(\beta_0)\} \rightarrow 1.$$

4 Projection Lemma

Lemma 1 (Growing sieve projection is negligible). *Under Assumptions 2–3,*

$$\|\mathbf{w}^c - \tilde{\mathbf{w}}\|_T = O_p(\sqrt{K/T}), \quad (10)$$

$$\max_{t \leq T} |w_t^c - \tilde{w}_t| = O_p(\zeta_K \sqrt{K/T}) = o_p(1), \quad (11)$$

$$T^{-1/2} \sum_{t=1}^T U_t(w_t^c - \tilde{w}_t) = O_p(K/\sqrt{T}) = o_p(1), \quad (12)$$

$$\|\mathbf{\Pi}_K \mathbf{U}\|_T = O_p(\sqrt{K/T}), \quad (13)$$

$$\max_{t \leq T} |(\mathbf{\Pi}_K \mathbf{U})_t| = O_p(\zeta_K \sqrt{K/T}) = o_p(1). \quad (14)$$

Proof. Because the sieve contains a constant and $\mathbf{1}'_T \mathbf{R} = \mathbf{0}$,

$$\mathbf{w}^c = \tilde{\mathbf{w}} - \mathbf{R} \hat{\boldsymbol{\delta}}_K, \quad \hat{\boldsymbol{\delta}}_K = (\mathbf{R}' \mathbf{R})^{-1} \mathbf{R}' \tilde{\mathbf{w}}.$$

By the Gram condition and (7), $\|\hat{\boldsymbol{\delta}}_K\| = O_p(\sqrt{K/T})$. Hence

$$\|\mathbf{R} \hat{\boldsymbol{\delta}}_K\|_T^2 = \hat{\boldsymbol{\delta}}_K' (T^{-1} \mathbf{R}' \mathbf{R}) \hat{\boldsymbol{\delta}}_K = O_p(K/T),$$

which proves (10). The maximum bound follows from $\max_t |\mathbf{R}'_t \hat{\boldsymbol{\delta}}_K| \leq \zeta_K \|\hat{\boldsymbol{\delta}}_K\|$. Also,

$$T^{-1/2} \mathbf{U}'(\mathbf{w}^c - \tilde{\mathbf{w}}) = -(T^{-1/2} \mathbf{R}' \mathbf{U})' \hat{\boldsymbol{\delta}}_K = O_p(\sqrt{K}) O_p(\sqrt{K/T}) = O_p(K/\sqrt{T}).$$

For the projection of $\mathbf{U}$, let $\hat{\boldsymbol{\gamma}}_U = (\mathbf{P}' \mathbf{P})^{-1} \mathbf{P}' \mathbf{U}$. Assumption 2 implies $\|\hat{\boldsymbol{\gamma}}_U\| = O_p(\sqrt{K/T})$, and the same quadratic-form and maximum-row arguments give (13)–(14). $\square$

5 Feasible-Oracle Equivalence

Lemma 2 (Score and denominator equivalence). *Under Assumptions 1–3, under $H_0 : \beta = \beta_0$,*

$$S_T := T^{-1/2} \sum_{t=1}^T \hat{Z}_t(\beta_0) = T^{-1/2} \sum_{t=1}^T U_t \tilde{w}_t + o_p(1), \quad (15)$$

$$\hat{V}_T := T^{-1} \sum_{t=1}^T \hat{Z}_t(\beta_0)^2 = T^{-1} \sum_{t=1}^T U_t^2 \tilde{w}_t^2 + o_p(1), \quad (16)$$

$$\max_{t \leq T} |\hat{Z}_t(\beta_0)| = o_p(\sqrt{T}). \quad (17)$$

Consequently, $\hat{V}_T \rightarrow_p \eta^2$.

Proof. Under the null,

$$\mathbf{Y} - \beta_0 \mathbf{X}_{lag} = \mathbf{P} \mathbf{c}_K + \mathbf{r}_K + \mathbf{U}, \quad \hat{\mathbf{u}}(\beta_0) = \mathbf{M}_K (\mathbf{U} + \mathbf{r}_K).$$

Using symmetry and idempotence of $\mathbf{M}_K$,

$$S_T = T^{-1/2} (\mathbf{U} + \mathbf{r}_K)' \mathbf{w}^c.$$

The term $T^{-1/2}\mathbf{U}'\mathbf{w}^c$ equals $T^{-1/2}\mathbf{U}'\tilde{\mathbf{w}} + o_p(1)$ by Lemma 1. The approximation term satisfies

$$|T^{-1/2}\mathbf{r}'_K\mathbf{w}^c| \leq \sqrt{T}\|\mathbf{r}_K\|_T\|\mathbf{w}^c\|_T \leq \sqrt{T}\|\mathbf{r}_K\|_T\|\mathbf{w}\|_T = o_p(1),$$

which proves (15).

Define the feasible-oracle residuals

$$\Delta_{u,t} = \hat{u}_t(\beta_0) - U_t, \quad \Delta_{w,t} = w_t^c - \tilde{w}_t, \quad \Delta_{Z,t} = \hat{Z}_t(\beta_0) - U_t\tilde{w}_t.$$

Since $\Delta_u = -\Pi_K\mathbf{U} + \mathbf{M}_K\mathbf{r}_K$, Lemma 1 and Assumption 3 give

$$\|\Delta_u\|_T = O_p(\sqrt{K/T} + a_K) = o_p(1).$$

Furthermore $\max_t |\Delta_{w,t}| = o_p(1)$, $\max_t |w_t^c| = O_p(1)$, and $T^{-1}\sum_t U_t^2 = O_p(1)$. Therefore

$$\Delta_{Z,t} = \Delta_{u,t}w_t^c + U_t\Delta_{w,t}$$

implies

$$\|\Delta_Z\|_T^2 \leq 2\max_t |w_t^c|^2 \|\Delta_u\|_T^2 + 2\max_t |\Delta_{w,t}|^2 T^{-1} \sum_{t=1}^T U_t^2 = o_p(1).$$

Let $V_T = T^{-1}\sum_t U_t^2 \tilde{w}_t^2$. Then

$$\hat{V}_T - V_T = \frac{2}{T} \sum_{t=1}^T U_t \tilde{w}_t \Delta_{Z,t} + \|\Delta_Z\|_T^2,$$

and Cauchy-Schwarz gives

$$|\hat{V}_T - V_T| \leq 2V_T^{1/2}\|\Delta_Z\|_T + \|\Delta_Z\|_T^2 = o_p(1),$$

because $V_T = O_p(1)$ by Assumption 1. This proves (16); $\hat{V}_T \rightarrow_p \eta^2$ follows from (6).

Finally, the uniform moment bound implies $\max_t |U_t| = o_p(\sqrt{T})$. Combining this with (14), $\|\mathbf{M}_K\mathbf{r}_K\|_\infty = O(a_K)$, and $\max_t |w_t^c| = O_p(1)$ proves (17). $\square$

6 Wilks Theorem

Theorem 1 (Sieve-orthogonalized mean EL). *Suppose Assumptions 1–4 hold. Under $H_0 : \beta = \beta_0$,*

$$\ell_T(\beta_0) \xrightarrow{d} \chi_1^2.$$

Proof. By Assumption 1 and Lemma 2,

$$S_T \xrightarrow{\text{st}} MN(0, \eta^2), \quad \hat{V}_T \xrightarrow{p} \eta^2.$$

Thus $S_T^2/\hat{V}_T \xrightarrow{d} \chi_1^2$. The convex-hull condition, nondegeneracy, and $\max_t |\hat{Z}_t(\beta_0)| = o_p(\sqrt{T})$ imply the standard scalar-EL bounds

$$\hat{\lambda} = O_p(T^{-1/2}), \quad \max_t |\hat{\lambda}\hat{Z}_t(\beta_0)| = o_p(1).$$

Expanding (5) gives

$$\hat{\lambda} = \frac{\sum_t \hat{Z}_t(\beta_0)}{\sum_t \hat{Z}_t(\beta_0)^2} + o_p(T^{-1/2}).$$

A second-order expansion of (4) yields

$$\ell_T(\beta_0) = \frac{\{T^{-1/2}\sum_t \hat{Z}_t(\beta_0)\}^2}{T^{-1}\sum_t \hat{Z}_t(\beta_0)^2} + o_p(1) = \frac{S_T^2}{\hat{V}_T} + o_p(1).$$

The conclusion follows. $\square$

7 Technical Remarks

Remark 1 (What is imported from CCHT). *The oracle convergence $A_T \xrightarrow{\text{st}} MN(0, \eta^2)$ and $V_T \rightarrow_p \eta^2$ is imported as the CCHT mean-score theorem under the enlarged filtration including W_t . The new work here is the perturbation argument proving $S_T = A_T + o_p(1)$ and $\hat{V}_T = V_T + o_p(1)$.*

Remark 2 (Why this is a mean theorem). *For quantile scores, the fitted nuisance error enters through a nonlinear indicator. The algebraic identity $\mathbf{P}'\mathbf{w}^c = \mathbf{0}$ no longer cancels the first-stage effect directly. A quantile extension would require a density-weighted orthogonalization or additional nonparametric expansion arguments.*